



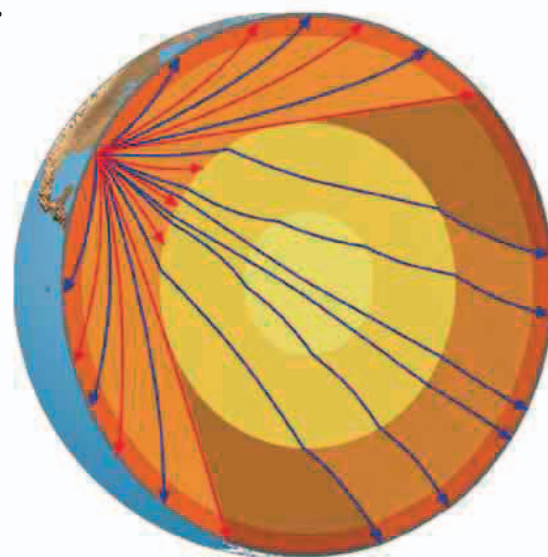
PRESSURE AND HEAT

The rock deep within the planet is far hotter than molten lava, but it is kept solid by intense pressure. It's similar to a bottle of soda. As long as the cap stays on, high pressure inside the bottle keeps it liquid. But if you release the pressure, some of the liquid turns to gas that fizzes out of the top.



EARTHQUAKE CLUES

We only know what lies beneath Earth's crust by studying the seismic waves that radiate from earthquakes. These waves pass at various speeds through the planet, but can be blocked by its different layers. The science of interpreting these earthquake clues is called seismology.



▲ SEISMIC WAVES

Some waves (blue) can pass through the core, but others (red) cannot and must go around it.

The upper mantle is kept moving by heat currents rising from near Earth's core.

Earth is a mass of searingly hot rock and metal with a thin, brittle shell.

FAST FACTS

Diameter of the inner core 1,516 miles (2,440 km)
Depth of the outer core 1,408 miles (2,266 km)
Depth of the mantle 1,800 miles (2,900 km)
Depth of the crust 3–43 miles (5–70 km)